SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

18. Crowd Monitoring System

Requirements:

- **Stable detection in dense crowds**
- **Moderate accuracy acceptable**

Options:
Optical Flow: unstable in dense scenes
Motion Models: stable

Compare and evaluate. Justify your decision.

SOLUTION:

Method	Dense Crowd Performance	Accuracy	Stability
Optical Flow	Poor	Moderate	Unstable
Motion Models	Good	Moderate	Stable

Comparison

- Optical Flow: Works well when people are clearly separated, but becomes unstable in dense crowds because people overlap and their movements are difficult to distinguish.
- Motion Models: Can handle complex and overlapping movements better, making detection more stable in crowded environments.

Best Choice: Motion Models

Motion Models should be selected because the main requirement is stable detection in dense crowds. Since moderate accuracy is acceptable, there is no need to choose a more complex method solely for higher accuracy.

Conclusion

Motion Models are the better choice because they provide stable and reliable detection in dense crowd conditions, which directly satisfies the system requirements.

30. Motion Animation System

Requirements:

- **Smooth motion**
- **Real-time rendering**

Options:
Motion Models: realistic, slow
Motion Estimation: fast, less realistic

Compare and evaluate design approach. Justify your decision.

SOLUTION:

Option	Motion Quality	Speed	Evaluation
Motion Models	Realistic and smooth	Slow	High visual quality but may cause delays in real-time rendering
Motion Estimation	Less realistic	Fast	Suitable for real-time applications but may reduce visual quality

Comparison

- Motion Models: Produce more realistic and smooth motion, but their slow processing can cause lag, which is undesirable for real-time animation.
- Motion Estimation: Provides fast processing, making it suitable for real-time rendering, although the motion may appear less realistic.

Best Design Approach: Motion Estimation

Motion Estimation should be selected because real-time rendering is a key requirement. Smoothness in real-time depends not only on realism but also on maintaining a high and consistent frame rate.

If additional processing power is available, a hybrid approach can be used: Motion Estimation for real-time updates and Motion Models for important scenes where higher realism is required.

Conclusion

For a real-time animation system, Motion Estimation is the better choice because it prioritizes fast rendering and responsive motion. The slight reduction in realism is acceptable when real-time performance is the main requirement.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- **Prioritize safety (accuracy > 90%)**
- **Maintain response time < 50 ms**

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

SOLUTION:

Method	Accuracy	Response Time	Meets Accuracy?	Meets Time?	Evaluation
A	88%	30 ms	❌ No	✅ Yes	Fast but not accurate enough for safety
B	93%	70 ms	✅ Yes	❌ No	Accurate but too slow
C	91%	45 ms	✅ Yes	✅ Yes	Satisfies both constraints

Constraint-Based Evaluation

- Method A: Fails the accuracy requirement because 88% < 90%. It should not be selected for a safety-critical autonomous vehicle.
- Method B: Meets the accuracy requirement but fails the response-time requirement because 70 ms > 50 ms. The delay could affect vehicle reaction time.
- Method C: Meets both requirements: 91% accuracy > 90% and 45 ms < 50 ms.

Best Choice: Method C

Method C should be selected because it is the only method that satisfies both safety and real-time constraints.

Conclusion:
Using constraint-based reasoning, Method C is the clear choice. Although Method B has higher accuracy, its response time violates the strict latency constraint. Method A is faster but does not meet the required safety accuracy. Therefore, 91% accuracy with 45 ms response time makes Method C the most suitable for deployment.

26. Video Surveillance Upgrade

Constraints:

- **Improve accuracy**
- **Maintain same latency**

Options:
Model A: 85%, 50 ms
Model B: 90%, 50 ms

Compare and evaluate. Justify your decision.

SOLUTION:

Model	Accuracy	Latency	Evaluation
Model A	85%	50 ms	Lower accuracy with the same latency

Model B	90%	50 ms	Higher accuracy with no increase in latency

Comparison

- Model A provides 85% accuracy with a latency of 50 ms.
- Model B improves accuracy to 90% while maintaining the same 50 ms latency.
- Since the main requirement is to improve accuracy without increasing latency, Model B directly satisfies both requirements.

Best Choice: Model B

Model B should be selected because it increases accuracy by 5 percentage points while keeping latency unchanged at 50 ms.

Conclusion:
Model B is the better upgrade because it provides higher detection accuracy without sacrificing real-time performance.

15. Face Recognition System Design

Constraints:

- **Accuracy ≥ 90%**
- **Works in low light**
- **Moderate computation**

Options:
Traditional: 82%
Deep Learning: 93%
Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

SOLUTION:

Method	Accuracy	Low-Light Performance	Computation	Evaluation
Traditional	82%	Limited	Low	Fails the accuracy requirement

Deep Learning	93%	Good	High	Meets accuracy but may require more computation
Hybrid	89%	Moderate	Moderate	Efficient but fails the accuracy requirement

Comparison

- Traditional method: Has only 82% accuracy, so it does not satisfy the minimum requirement of 90%.
- Deep Learning: Achieves 93% accuracy, satisfying the accuracy constraint. It can also perform well under low-light conditions, especially when trained with suitable low-light images. However, it requires relatively higher computational resources.
- Hybrid method: Requires moderate computation, but its 89% accuracy is below the required 90%, so it cannot be selected.

Best Choice: Deep Learning

Deep Learning should be selected because it is the only option that satisfies the accuracy requirement of ≥90% and can provide strong performance in low-light conditions.

Although its computation is higher, the requirement says moderate computation, not extremely low computation. Therefore, a lightweight or optimized deep-learning model can be deployed to reduce computational requirements.

Conclusion

Deep Learning is the best choice because it provides 93% accuracy, supports reliable low-light recognition, and can be optimized for moderate computational resources. The traditional and hybrid methods fail the minimum accuracy constraint.

20. Smart Parking System

Requirements:

- **Detect vehicles accurately**
- **Low computational cost**

Options:
Viola-Jones: low cost, low accuracy
YOLO: moderate cost, high accuracy Compare and evaluate. Justify your decision.

SOLUTION:

Method	Accuracy	Computational Cost	Evaluation
Viola-Jones	Low	Low	Efficient but may miss vehicles or produce incorrect detections
YOLO	High	Moderate	More accurate but requires more computational resources

Comparison

- Viola-Jones: Has low computational cost, making it suitable for low-power systems. However, its lower accuracy can reduce the reliability of vehicle detection.
- YOLO: Provides high detection accuracy and is more reliable for identifying vehicles. Although it requires moderate computational resources, it offers better detection performance.

Best Choice: YOLO

YOLO should be selected if accurate vehicle detection is the main priority. The moderate computational cost can be managed using a lightweight YOLO model or edge-device optimization.

If the parking system has extremely limited hardware and cost is more important than detection accuracy, Viola-Jones can be considered.

Conclusion

YOLO is the better overall choice because it provides high vehicle-detection accuracy while requiring only moderate computational resources. For a smart parking system, reliable detection is important for correctly identifying available and occupied parking spaces.

14. Motion Tracking in Sports Analytics

Requirements:

- **Accurate motion tracking**
- **Real-time performance**

Options:
Optical Flow: accurate, slow
Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

SOLUTION:

Method	Accuracy	Speed	Evaluation
Optical Flow	High	Slow	Provides accurate tracking but may not meet real-time requirements
Motion Estimation	Lower	Fast	Suitable for real-time tracking but sacrifices some accuracy

Comparison

- Optical Flow: Provides accurate motion tracking, which is useful for detailed sports analysis. However, its slower processing may cause delays during real-time applications.
- Motion Estimation: Provides fast tracking, making it suitable for real-time performance. However, its lower accuracy may reduce the quality of detailed analysis.

Constraint-Based Evaluation

The system has two requirements: accurate tracking and real-time performance. Neither option completely satisfies both:

- Optical Flow → satisfies accuracy but fails real-time speed.
- Motion Estimation → satisfies real-time speed but has lower accuracy.

Best Choice: Motion Estimation

Motion Estimation is the more suitable choice when real-time performance is a strict constraint. In sports analytics, immediate tracking can be more important than maximum accuracy. Its accuracy can also be improved through optimization or combining it with more accurate methods when necessary.

Conclusion

Motion Estimation should be selected because it satisfies the real-time performance requirement while providing reasonable motion tracking. If highly precise post-match analysis is more important than real-time processing, Optical Flow would be preferable.

11. Traffic Surveillance Model Selection

A traffic monitoring system must:

- **Accuracy ≥ 90%**
- **Latency ≤ 80 ms**
- **Works under varying lighting**

Options:
Method A: 88%, 60 ms
Method B: 91%, 85 ms
Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

SOLUTION:

The system has three constraints:

1. Accuracy ≥ 90%
2. Latency ≤ 80 ms
3. Works under varying lighting conditions

Method	Accuracy	Latency	Accuracy Constraint	Latency Constraint	Evaluation
A	88%	60 ms	❌ Fail	✅ Pass	Fast, but accuracy is insufficient
B	91%	85 ms	✅ Pass	❌ Fail	Accurate, but too slow
C	92%	75 ms	✅ Pass	✅ Pass	Best overall option

Comparison

- Method A: Has good latency (60 ms) but fails the accuracy requirement because 88% < 90%.

- Method B: Meets the accuracy requirement (91%) but fails the latency requirement because 85 ms > 80 ms.
- Method C: Meets both requirements with 92% accuracy and 75 ms latency.

Best Choice: Method C

Method C should be selected because it is the only method that satisfies both numerical constraints. It also needs to be verified under varying lighting conditions before deployment.

Conclusion

Method C is the most suitable choice, providing 92% accuracy and 75 ms latency. Therefore, it meets the accuracy and response-time requirements while leaving a 5 ms latency margin. For final deployment, its performance under different lighting conditions should also be tested.